

## BIOLOGY - THE STUDY OF LIFE

# Module map

- Module 01: Biology - The Study of Life
- Connected lab:  
lab-01\_measurement-methods.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 01

Biology - The Study of Life

1

What counts as life

2

Science as a way of testing

3

Organization from cells to ecosystems

4

Evidence, variables, and measurement

5

How biological explanations improve

### Lab evidence

lab-01\_measurement-methods.md

## BIOLOGY - THE STUDY OF LIFE

# Learning objectives

- Distinguish living systems from nonliving examples using multiple characteristics.
- Separate hypotheses, predictions, observations, and conclusions in a scientific test.
- Place biological examples at the correct scale of organization.
- Identify variables and evidence in a simple investigation.
- Revise a biological explanation when new evidence appears.

**OBJECTIVE 1**

Distinguish living systems from nonliving examples using multiple characteristics.

**OBJECTIVE 2**

Separate hypotheses, predictions, observations, and conclusions in a scientific test.

**OBJECTIVE 3**

Place biological examples at the correct scale of organization.

**OBJECTIVE 4**

Identify variables and evidence in a simple investigation.

**OBJECTIVE 5**

Revise a biological explanation when new evidence appears.

## BIOLOGY - THE STUDY OF LIFE

# Topic sequence

- What counts as life: Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.
- Science as a way of testing: Scientific claims are tested with observations, comparisons, variables, and repeatable reasoning.
- Organization from cells to ecosystems: Biological systems are organized across scales from molecules to biosphere.
- Evidence, variables, and measurement: Data become useful when measurements are tied to a clear question and source of error.
- How biological explanations improve: Strong explanations connect evidence to a claim and stay open to revision.

01

## What counts as life

Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.

02

## Science as a way of testing

Scientific claims are tested with observations, comparisons, variables, and repeatable reasoning.

03

## Organization from cells to ecosystems

Biological systems are organized across scales from molecules to biosphere.

04

## Evidence, variables, and measurement

Data become useful when measurements are tied to a clear question and source of error.

05

## How biological explanations improve

Strong explanations connect evidence to a claim and stay open to revision.

## BIOLOGY - THE STUDY OF LIFE

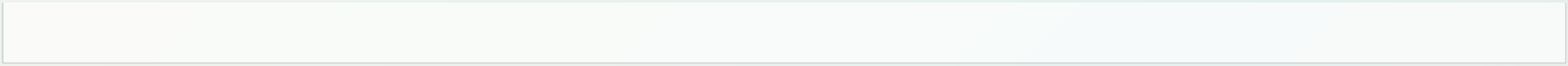
# Module 01: Biology - The Study of Life Evidence Map

- Module focus: Biology - The Study of Life
- Central claim: Biology - The Study of Life explains life criteria with measurement-based scientific explanation.
- Key nodes to connect: What counts as life, Science as a way of testing, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection:  
lab-01\_measurement-methods.md  
supplies an evidence surface for the map.

## BIOLOGY - THE STUDY OF LIFE

# Module 01: Biology - The Study of Life Reasoning Sequence

- Module focus: Biology - The Study of Life
- Trace the model: What counts as life  
-> Science as a way of testing ->  
Organization from cells to  
ecosystems -> Evidence, variables,  
and measurement
- Inputs become outputs: What counts  
as life, Science as a way of testing ->  
Distinguish living systems from  
nonliving examples using multiple  
characteristics., Separate hypotheses,  
predictions, observations, and  
conclusions in a scientific test.
- Feedback check:  
lab-01\_measurement-methods.md  
evidence checks whether students



## BIOLOGY - THE STUDY OF LIFE

# Terms as evidence handles

- Cell: Basic unit of life bounded by a membrane.
- Homeostasis: Regulation that keeps internal conditions within a workable range.
- Metabolism: All chemical reactions that move matter and energy through life.
- Hypothesis: Testable explanation for an observed pattern.
- Variable: Factor that can change in an investigation.
- Evidence: Observation or measurement used to support or challenge a claim.

**CELL**

Basic unit of life bounded by a membrane.

**HOMEOSTASIS**

Regulation that keeps internal conditions within a workable range.

**METABOLISM**

All chemical reactions that move matter and energy through life.

**HYPOTHESIS**

Testable explanation for an observed pattern.

**VARIABLE**

Factor that can change in an investigation.

**EVIDENCE**

Observation or measurement used to support or challenge a claim.

## BIOLOGY - THE STUDY OF LIFE

# Lab connection

- Lab file: lab-01\_measurement-methods.md
- Lab evidence should test or illustrate:  
Organization from cells to ecosystems
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**

What counts as life

**EVIDENCE**

lab-01\_measurement-methods.md

**CLAIM**

Distinguish living systems from nonliving examples using multiple characteristics.

## BIOLOGY - THE STUDY OF LIFE

# Contrast check

- Do not stop at: Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.
- Stronger explanation: Strong explanations connect evidence to a claim and stay open to revision.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

Life is recognized by shared properties: cells, energy use, response, reproduction, regulation, and evolution.

**DEEPER**

Strong explanations connect evidence to a claim and stay open to revision.

## BIOLOGY - THE STUDY OF LIFE

# Module 01: Biology - The Study of Life Retrieval and Lab Check

- Module focus: Biology - The Study of Life
- Retrieval prompt: Which characteristics together make something count as alive?
- Answer check: Distinguish living systems from nonliving examples using multiple characteristics.
- Required terms: Cell, Homeostasis, Metabolism
- Lab connection:  
lab-01\_measurement-methods.md;  
lab-01\_measurement-methods.md  
supplies evidence for measurement choices, repeated observations, and error sources.

## BIOLOGY - THE STUDY OF LIFE

# Practice quiz bridge

- 1. A student sees frost crystals slowly "grow" on a cold window overnight. Why don't biologists count this as life?
- 2. To test whether fertilizer speeds plant growth, a student changes the amount of fertilizer given to each plant. The amount of fertilizer is the:
- 3. Which list places levels of biological organization from smallest to largest?
- 4. A hypothesis survives many careful tests over many years. What should happen to it?

**QUESTION 1****Answer first**

A student sees frost crystals slowly "grow" on a cold window overnight. Why don't biologists count this as life?

**QUESTION 2****Answer first**

To test whether fertilizer speeds plant growth, a student changes the amount of fertilizer given to each plant. The amount of fertilizer is the:

**QUESTION 3****Answer first**

Which list places levels of biological organization from smallest to largest?

**QUESTION 4****Answer first**

A hypothesis survives many careful tests over many years. What should happen to it?

## BIOLOGY - THE STUDY OF LIFE

# Synthesis and exit ticket

- Exit claim: explain what counts as life using evidence from lab-01\_measurement-methods.md.
- Must include: Cell, Homeostasis, Metabolism.
- Revision prompt: what evidence would change your explanation?

**CLAIM**

What counts as life

**EVIDENCE**

lab-01\_measurement-methods.md

**REVISION**

What would change your mind?